RACK DESIGNER

User Manual

Version 0.1.21

Stage Tools

stage-tools.github.io/stagetools

1. Introduction

Rack Designer is a rack layout application designed for live events, touring productions, permanent installations, and broadcast environments. It lets you build and document equipment racks, manage a reusable device library, and export rider-ready PDF reports.

1.1 What Rack Designer Does

- Design multi-rack layouts visually by dragging equipment from a library into rack slots.
- Manage a persistent device library with technical specifications (power, weight, dimensions).
- Support tray configurations — place multiple half-width or sub-rack-width devices side by side within a single rack row.
- Export PDF documentation including rack diagrams, device lists, power and weight totals, and project summaries.
- Store multiple projects independently, each with its own metadata and rack configuration.

1.2 System Requirements

Requirement	Details
Operating System	Windows 10 / 11 (64-bit)
Display	1280 × 800 pixels minimum, higher resolution recommended
Storage	~50 MB for the application; additional space for projects and library

1.3 Application Layout

The application is organised into five top-level sections accessible from the navigation bar at the top of the window:

Section	Purpose
PROJECT	Create, open, save, and delete projects. Edit project metadata.
DESIGNER	Visual rack layout editor. Add racks, drag devices, reorder slots.
EXPORT	Configure and generate PDF documentation.
LIBRARY	Manage the device library: add, edit, duplicate, delete, import, export.
ABOUT	Version information and links.

> ***Note:*** *The navigation bar also shows the current project name and a SAVE button. An amber dot (●) next to the project name indicates unsaved changes.*

2. Projects

A project is the top-level container for your work. Each project holds metadata and one or more racks. Projects are saved to disk as .rackproj files.

2.1 Creating a New Project

Click PROJECT in the navigation bar, then click NEW PROJECT in the sidebar. A blank project is created in memory. Fill in the fields in the main area and click SAVE (or press Ctrl+S) to write it to disk for the first time.

> ***Note:*** *A new project does not appear in the Recent Projects list until it has been saved at least once.*

2.2 Project Metadata Fields

Field	Description
Project name	Displayed in the navigation bar, window title, and exported PDFs.
Project number	Free-text reference (e.g. job or commission number). Exported as the Client field in PDFs.
Venue	Location or venue name. Appears in PDF exports.
Date	Free-text date (e.g. 2025-06-01). Appears in PDF exports.
Notes	Free-text notes. Appear in PDF exports when the Notes option is enabled.

2.3 Saving a Project

- Click SAVE in the navigation bar or the Project section footer, or press Ctrl+S.
- The first time a project is saved, a .rackproj file is created automatically in the application data directory.
- If a project with the same name already exists on disk, a numeric suffix is appended automatically (e.g. "My Show (2)").
- To save to a specific location, click SAVE AS… in the Project section footer.

2.4 Opening an Existing Project

- Select a project from the Recent Projects list in the left sidebar of the PROJECT section.
- Click Ctrl+O to browse for a .rackproj file anywhere on disk.

> ***Note:*** *If the current project has unsaved changes, a dialog will ask whether to save before loading another project. Choose Save, Discard, or Cancel.*

2.5 Deleting a Project

Select a project in the Recent Projects list and click DELETE PROJECT (or the DELETE button in the footer). Confirm the action in the dialog. This permanently removes the .rackproj file from disk and cannot be undone.

> ***Note:*** *Deleting a project does not affect any library entries — only the project file is removed.*

3. Designer

The Designer is the main workspace. It is divided into a resizable sidebar on the left and the rack canvas on the right. The sidebar has two tabs: RACKS and LIBRARY.

3.1 Racks Tab

The Racks tab lists all racks in the current project. Click a rack to select it. The selected rack is the target for the double-click action in the Library tab and for the custom slot section. You can reorder racks by dragging them within the list — the order is reflected immediately on the canvas.

When a slot is selected on the canvas, a slot information panel appears below the rack list, showing the device name, category, physical dimensions, power draw, weight, and any notes attached to the slot.

3.1.1 Adding a Rack

Click + ADD RACK at the bottom of the Racks tab. In the dialog:

- Enter a name for the rack.
- Optionally select a common size preset (4U to 48U).
- Set the size in U (rack units). Range: 1–60.
- Set the number of bays (1, 2, or 3 side-by-side vertical sections within the same rack enclosure).

Note: *Bays allow you to represent multi-bay physical rack cases. Each bay is displayed as an independent column on the canvas at the same U height.*

3.1.2 Editing a Rack

Select a rack in the list and click EDIT. You can rename the rack and resize it. If you reduce the rack size, the application checks whether any slots in the rows that would be removed are occupied and refuses the change with a list of the conflicting devices.

Note: *When a rack is enlarged, existing slots shift downward to make room at the top. When it is reduced, existing slots shift upward.*

3.1.3 Removing a Rack

Select a rack in the list and click REMOVE. Confirm in the dialog. The rack and all its slots are permanently removed from the project. The library is not affected.

3.2 Library Tab

The Library tab lists all devices in your device library. Filter by category using the drop-down and search by name, manufacturer, model, or description using the search field.

3.2.1 Adding a Device to a Rack

There are two ways to place a library device into a rack:

- Drag: Click and hold a device in the library list, drag it onto the rack canvas, and release at the desired row. A highlight shows the target row under the cursor.
- Double-click: Double-click a device in the library list. It is placed at the currently selected unit on the canvas, or at the first available position if no unit is selected.

Tip: *Click an empty unit on the canvas first to select it, then double-click a device in the library to place it at that exact position.*

3.2.2 Tray Placement

Devices that are not rack-native (i.e. narrower than a full 19" rack width, 482 mm) can be placed side by side in the same row to form a tray. When you drag such a device onto an existing slot that already contains another sub-width device and there is enough horizontal space remaining, the two devices are automatically combined into a tray.

> ***Note:*** *Tray capacity is checked in millimetres. The standard rack inner width is 482 mm. Rack Designer refuses placement if the combined width of all tray members would exceed this limit.*

3.3 Custom Slots

At the bottom of the Library tab is a Custom Slot section. Enter a label, set a height in U, then click "Add to selected rack". If a unit is selected on the canvas, the custom slot is inserted at that exact position (fails if the unit is occupied). If no unit is selected, the slot is placed at the first available position.

Custom slots can be renamed and resized after insertion via right-click → Edit slot…

3.4 The Rack Canvas

The canvas displays all project racks side by side. Each rack is a vertical column of U-height rows. U1 is at the top. U number labels are shown to the left of the leftmost bay.

3.4.1 Selecting a Slot or Unit

Click an occupied slot to select it (amber border). Click an empty unit to select that position (subtle amber highlight). The selected position is used as the insertion target for double-click and custom slot operations. Selecting a unit in one rack automatically clears the selection in all other racks.

3.4.2 Navigating the Canvas

The workspace behaves like a cartesian plane:

Action	Result
Scroll wheel	Scroll vertically
Shift + Scroll wheel	Scroll horizontally
Ctrl + Scroll wheel	Zoom in / out (all racks simultaneously)
Ctrl + + / Ctrl + −	Zoom in / out via keyboard
Ctrl + 0	Fit all racks to the current window
FIT button (toolbar)	Same as Ctrl + 0
Right-click → Fit this rack	Scale to show the clicked rack at full height
Right-click → Fit all racks	Scale to show all racks at full height
Right-click → Reset zoom	Return to 100% zoom

3.4.3 Reordering Slots by Drag

Click and hold a slot on the canvas, then drag it up or down within the same bay. The slot cannot overlap another slot. Release to confirm the new position.

3.4.4 Context Menu

Right-click any slot on the canvas to open the context menu:

Action	Description
Edit notes…	Open a dialog to enter free-text notes for this slot. Slots with notes display an amber left-edge bar.
Edit slot…	Edit the slot name and U height. For tray slots (multiple devices sharing one row), the name field is not shown — each device inside a tray keeps its own name from the library.
Move up ↑	Move the slot one unit upward (if space allows).
Move down ↓	Move the slot one unit downward (if space allows).
Remove from rack	Remove the slot from the rack. The device definition remains in the library.
Remove device from tray…	For tray slots: a sub-menu lists each device individually for selective removal.
Copy (Ctrl+C)	Copy the selected slot to the clipboard.
Cut (Ctrl+X)	Cut the selected slot to the clipboard and remove it from the rack.
Paste (Ctrl+V)	Paste the clipboard slot at the first free position in the active bay.
Duplicate (Ctrl+D)	Duplicate the selected slot at the first available position in the same bay.
Delete (Del)	Remove the selected slot from the rack (confirmation dialog; Enter confirms).
Undo (Ctrl+Z)	Undo the last operation on the active rack (up to 10 levels).
Fit this rack to window	Scale the view to show this rack at full height.
Fit all racks to window	Scale the view to show all racks at full height.
Reset zoom (100%)	Return to 100% zoom.

Right-clicking on an empty row also offers: Insert blank panel at U[n], and the same zoom/fit actions.

3.4.5 Stats Bar

Below the canvas a stats bar shows the total used U, free U, number of bays (when more than one), total power draw in watts, and total weight in kg for the entire project.

3.4.6 Zoom Toolbar

Above the canvas, a compact toolbar shows the current zoom percentage and provides − / + / FIT buttons. Zoom actions always apply to all racks simultaneously.

4. Library Management

The device library is a persistent, project-independent database of equipment definitions. Changes to the library do not retroactively affect devices already placed in racks — each slot stores a snapshot of the device definition at the time of placement.

4.1 Device Fields

Field	Description
Name	Required. Display name of the device.
Manufacturer	Free-text manufacturer name.
Model	Free-text model identifier.
Description	Short description of the device.
Category	Functional category. Used for filtering and colour coding. Built-in categories: Video, Audio, Lighting, Network, Power, Other. Custom categories can be added.
Mounting	Rack-native (standard 19"): the device occupies a declared number of U. Non-rack-native: height in mm is used to compute U automatically.
Height (U)	Number of rack units occupied (rack-native devices only, range 1–20).
Dimensions W/H/D	Physical width, height, and depth in mm. Width below 482 mm marks the device as sub-width, making it eligible for tray placement. Height drives U calculation for non-rack-native devices.
Color	Custom colour swatch for display on the canvas. If left as "auto", the category default colour is used.
Power (W)	Power consumption in watts. Contributes to per-rack and per-project totals.
Weight (kg)	Device weight. Contributes to per-rack and per-project totals.
Notes	Free-text technical notes visible in the device library and optionally in PDF exports.
Images	Optional front and rear photos of the device, shown in the device edit form. Not included in CSV export or import (see 4.8).

4.2 Creating a Device

Click + NEW. A blank entry is created and selected. Fill in the fields in the right panel and click SAVE.

4.3 Editing a Device

Select a device in the list. Edit any fields in the right panel. Click SAVE to confirm. Changes apply to the library entry only — existing rack slots are not updated.

4.4 Duplicating a Device

Select a device and click DUPLICATE. A copy is created with " (copy)" appended to the name. Useful for creating variants (e.g. different power configurations of the same chassis).

4.5 Deleting a Device

Select a device and click DELETE. Confirm in the dialog. The device is removed from the library permanently. Projects that already contain this device are not affected.

4.6 Custom Categories

In addition to the six built-in categories, you can define your own. In the Category drop-down within the device form, select "+ Add category…" and enter a name. Custom categories are shared across all devices and projects.

4.7 Importing Devices

Click IMPORT in the Library toolbar. Supported formats:

JSON

A JSON array of device objects in the Rack Designer format (produced by the JSON export). All fields are preserved.

CSV

A semicolon-delimited file (comma is auto-detected if no semicolons are found). Expected column headers:

Column	Notes
Name	Required. If empty, Manufacturer and Model are combined as the name.
Manufacturer	Optional.
Model	Optional.
Description	Optional.
Category	Optional. Auto-capitalised. New categories are created automatically.
Power (W)	Optional. Accepts decimal values with comma or dot separator.
Height (U)	If present and valid, the device is treated as rack-native. If empty, H (mm) is used to compute U height; the device is treated as non-rack-native.
W	Physical width in mm. Optional.
H	Physical height in mm. Optional. Required for non-rack-native U calculation.
D	Physical depth in mm. Optional.
Weight (kg)	Optional.
Notes	Optional.

After parsing, a dialog asks whether to Merge or Replace:

- Merge: adds devices whose (name, category) pair is not already in the library. Duplicates are skipped.
- Replace: discards the existing library entirely and replaces it with the imported set.

4.8 Exporting Devices

Click EXPORT in the Library toolbar. A two-step dialog appears:

- Step 1 — Scope: export the entire library, a specific category, or only the currently selected device.
- Step 2 — Format: JSON (full fidelity) or CSV (semicolon-delimited, for editing in spreadsheet applications).

Note: *Exported JSON files can be shared with other Rack Designer users and imported directly.*

Note: *CSV export does not include device images (front/rear). Use JSON export to preserve them.*

5. PDF Export

Navigate to the EXPORT section to configure and generate a PDF document for the current project.

5.1 Rack Selection

The left column lists all racks with a checkbox. Only checked racks are included in the export. All racks are checked by default.

5.2 Content Options

Option	Description
Rack diagram	Draws a scaled graphical representation of each rack, showing device positions.
Use device colours	When enabled, the rack diagram prints with each device's assigned colour. When unchecked, the diagram uses a grayscale theme suitable for monochrome printing — different categories still appear in visibly different shades of gray for easy distinction.
Device list per rack	Appends a table below each rack listing devices with quantity, name, category, and optional columns.
Include category column	Adds a Category column to the device list table.
Power totals	Adds power consumption (watts) to the rack info line, device list, and project summary.
Weight totals	Adds weight (kg) to the rack info line, device list, and project summary.
Project summary	Appends a summary table at the end of the document covering all selected racks.
Slot notes	Adds a Notes column to the device list, showing any per-slot notes.

5.3 Generating the PDF

Click EXPORT PDF, choose a file path in the save dialog, and confirm. A success message appears when the file is written.

Note: *Rack diagrams scale automatically to fit the page if their natural height exceeds the available print area.*

5.4 PDF Structure

The generated PDF follows this structure:

- Document header: project name, client/project number, venue, date, generation date, and notes.
- Per-rack block: rack name, info line (size, used/free U; optional watts and kg), optional rack diagram, optional device list.
- Optional project summary table.
- Footer on every page: application credit line and page number.

5.5 PNG Export

In addition to PDF, individual rack diagrams can be exported as standalone PNG images. Click EXPORT RACKS AS PNG in the EXPORT section, then choose a destination folder. One PNG file is created for each checked rack (using the same rack selection checkboxes as PDF export), named after the rack.

> ***Note:*** *Unlike the PDF diagram, PNG export renders each rack at its full natural size — there is no page-fit scaling. This is useful for large or multi-bay racks that appear small in the PDF. Images are generated at 300 DPI and use the same "Use device colours" setting as PDF export. If two racks share the same name, a numeric suffix is added automatically so files are never overwritten.*

6. Preloaded Device Library

On first launch, Rack Designer seeds the library with a curated set of commonly used professional equipment across five categories.

6.1 Video

- Disguise: GX 1, GX 2, GX 3, GX 3+, VX 4+, 4x4 Pro, 2x2
- AV Stumpfl Pixera: PX Zero, PX One Gen2, PX Two Gen2, PX Four Gen2, PX Four RS Gen2
- Barco: ImagePro-4K, EX, S3, E2, E2 Gen2, Encore 3, PDS 4K
- Analog Way: Aquilon RS1–RS6, Ascender 16, 32, 48
- Novastar: MCTRL 4K, MCTRL R5, VX400/600/1000/2000 Pro, H2, H5, H9, H15, H20
- Pixelhue: 4K ViewPro, P10, P20, P20 DS, P80, P8
- Colorlight: Z3, Z6
- Lightware: MX2 8x8, 16x16, 24x24 HDMI; 16x16 DVI
- Blackmagic Design: Teranex AV; HyperDeck HD Mini, Plus, Pro, 4K; Videohub 10x10 to 80x80 12G; ATEM Constellation HD and 4K/8K series; MultiView 4 and 16; full Teranex Mini converter range

6.2 Audio

- Drawmer: DA6 Distribution Amplifier
- Tascam: US-16x8
- Focusrite: Scarlett 18i20 3rd Gen
- Yamaha: SWP1 Dante switches (SWP1-8MMF, SWP1-16MMF, SWP1-8S)

6.3 Lighting

- Luminex: LumiSplit 2.10, LumiNode 12

6.4 Network

- Luminex: Gigacore 10t, Gigacore 18t
- Netgear M4250 series: multiple PoE+ and PoE++ variants
- Netgear Prosafe: G716T, XS508M, XS708Ev2, XS512EM

6.5 Power

- Eaton UPS: 5PX 1500, 2200, 3000 (Gen1 and Gen2); 5P 1150, 1150 Gen2

Note: *The preloaded library is seeded only once, on first launch. All subsequent edits, additions, and deletions are preserved. Reinstalling the application will reseed the library only if it is empty.*

7. Keyboard Shortcuts

Shortcut	Action
Ctrl+S	Save the current project.
Ctrl+O	Open a project file from disk.
Ctrl+N	Create a new project.
Click (canvas)	Select a slot or empty unit. Click outside racks to deselect.
Click + Drag (slot)	Reorder a slot within its bay.
Right-click (canvas)	Open the context menu for the slot or empty row under the cursor.
Click + Drag (library)	Drag a device from the library onto the rack canvas.
Double-click (library)	Place a device at the selected unit, or first available position.
Ctrl+C	Copy the selected slot to the clipboard.
Ctrl+X	Cut the selected slot to the clipboard and remove it from the rack.
Ctrl+V	Paste the clipboard slot at the first free position in the active bay.
Ctrl+D	Duplicate the selected slot at the first available position.
Del	Remove the selected slot (confirmation dialog; Enter confirms).
Ctrl+Z	Undo the last operation on the active rack (up to 10 levels).
Ctrl + + / Ctrl + −	Zoom in / out (all racks).
Ctrl + 0	Fit all racks to the window.
Scroll wheel	Scroll the canvas vertically.
Shift + Scroll wheel	Scroll the canvas horizontally.

Ctrl + Scroll wheel	Zoom in / out (all racks).

8. Data Storage

Rack Designer stores all persistent data in the application data directory:

```
%APPDATA%\RackDesigner\
```

File / Folder	Contents
library.json	The complete device library.
custom_categories.json	User-defined custom categories.
projects	Folder containing all saved .rackproj project files.
crash.log	If an unexpected error occurs, details are logged here automatically, and a dialog explains what happened. The application continues running rather than closing without warning.

Project files (.rackproj) are plain JSON and can be backed up, version-controlled, or shared with other users. A shared .rackproj file contains a full snapshot of all device data and does not require the recipient to have the same library installed.

> ***Note:*** *There is no cloud sync or auto-save. Save your work regularly with Ctrl+S.*

9. Tips & Best Practices

9.1 Library Organisation

- Use a consistent naming convention: Manufacturer Model (e.g. "Novastar MCTRL R5").
- Always fill in Watts and Weight for equipment that carries electrical or structural significance.
- Use the Depth field to document rack depth requirements even if it is not visible on the canvas.
- For equipment available in multiple configurations, use DUPLICATE and adjust the copy rather than creating from scratch.

9.2 Rack Layout

- Reflect the physical placement: heavy equipment (UPS, amplifiers) at the bottom, network and processing at the top.
- Use blank panels (right-click on empty rows) to document intentionally empty U spaces.
- Use slot notes (right-click → Edit notes…) to attach cable labels, IP addresses, or configuration reminders.
- Multi-bay racks are useful for wide physical cases. Devices do not span bays.
- Click an empty unit before double-clicking a library device to place it at that exact position.
- Use Ctrl+0 or the FIT button to instantly fit all racks to the window after adding or removing racks.

9.3 PDF Export

- For rider and advance documentation: enable Rack diagram, Use device colours, Device list, Power totals, Weight totals, and Project summary.

- For a printed working copy: leave Use device colours unchecked for a clean white diagram readable on monochrome printers.
- Enable Slot notes to include per-slot annotations in the device list.

9.4 Project Files

- Keep a copy of the .rackproj file alongside your production documentation. The file is self-contained.
- Use SAVE AS… to create named revisions (e.g. MyShow_v2_preproduction.rackproj).

10. Troubleshooting

Issue	Solution
A device dragged from the library does not appear in the rack.	Ensure there is enough consecutive free U space in the target bay for the device height.
Double-clicking a library device inserts it in the wrong position.	Click the target unit on the canvas first to select it, then double-click the device.
Cannot reduce rack size.	Move or remove the listed devices before reducing the rack size.
A sub-width device is not combining into a tray.	Both devices must be non-rack-native with a Width value set. The combined width must not exceed 482 mm.
The PDF rack diagram is very small.	The diagram scales to fit one page. Very large racks produce proportionally smaller diagrams.
A project file cannot be opened.	Ensure the file ends in .rackproj and has not been corrupted. The file is plain JSON and can be inspected in a text editor.
Library changes are not reflected in existing rack slots.	This is by design. Slots store a snapshot of the device at placement time. Re-adding the device will use the current library definition.
Pressing Enter in a confirmation dialog does not confirm.	Update to v0.1.19 — Enter now confirms the default action in all dialogs.

11. Version History

Version	Highlights
0.1.17	Initial public release. Project management, visual rack designer, device library with 100+ preloaded devices, PDF export, tray placement, multi-bay rack support, CSV and JSON import/export.
0.1.18	Keyboard shortcuts for slot operations: Ctrl+C, Ctrl+X, Ctrl+V, Ctrl+D, Del. Undo stack per rack up to 10 levels (Ctrl+Z). All actions available via context menu with shortcut hints.
0.1.19	Scroll (vertical), Shift+Scroll (horizontal), Ctrl+Scroll / Ctrl+± (zoom all racks), Ctrl+0 / FIT (fit all). Zoom toolbar in canvas info bar. Rack list drag-to-reorder synced to canvas. Unit selection on canvas: click empty unit to target insertion position. Double-click library device and Add Custom Slot insert at selected unit. Edit slot dialog (name and U height). Enter confirms deletion dialogs. PDF rack diagrams use a white/light theme by default; "Use device colours" enables colour output. Interface brightness raised. Context menu fit/zoom actions available on occupied slots too.
0.1.20	PNG export for rack diagrams — export each rack as a standalone image at full natural size (no page-fit scaling) and 300 DPI, alongside the existing PDF export. Device images (front/rear photos) now save and load correctly in the device library. Grayscale PDF export ("Use device colours" unchecked) now visually distinguishes categories by shade instead of rendering everything the same neutral color. Unexpected errors are now logged to crash.log and shown in a dialog instead of the application closing without warning. Numerous reliability improvements to project saving/loading, device library import/export, and rack editing.
0.1.21	Released as free software under the GNU AGPL v3 or later. Licence and no-warranty notices in About, source code repository, THIRD-PARTY-NOTICES.txt and LICENSE installed with the program.

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Source code. The complete corresponding source code of this program, including that of every released version, is available at https://github.com/Stage-Tools/rackdesigner-src. If you modify the program and let users interact with it over a network, the AGPL requires you to offer them the source of your modified version.

Third-party components. Rack Designer includes PyQt6 and Qt, PyMuPDF and MuPDF, reportlab, Pillow and Python. Their licences are listed in THIRD-PARTY-NOTICES.txt, installed in the program folder.

Rack Designer is developed by Riccardo Nessi — Stage Tools

stage-tools.github.io/stagetools